

Note 2.4 Sequences and Summations

Introduction

In this section we will learn about the sequences and its use.

Sequences

Definition of Sequences

A **sequence** is a function from a subset of the set of integers (usually either the set $\{0, 1, \dots\}$ or the set $\{1, 2, \dots\}$) to a set S . We use the notation a_n to denote the image of the integer n . We call a_n a term of the sequence.

Definition of Geometric Progression

A **geometric progression** is a sequence of the form

$$a, ar, ar^2, ar^3, \dots, ar^n, \dots$$

where the **initial term** a and the **common ratio** r are real numbers.

Remarks

A geometric progression is a **discrete analogue** of the exponential function $f(x) = ar^x$.

Definition of Arithmetic Progression

An **arithmetic progression** is a sequence of the form

$$a, a + d, a + 2d, \dots, a + nd, \dots$$

where the **initial term** a and the **common difference** d are real numbers.

Remarks

An arithmetic progression is a discrete analogue of the linear function $f(x) = dx + a$.

Sequences in Computer Science

In the computer science, the finite sequences are also called **strings**. It is also denoted by

$$a_1 a_2 \dots a_n$$

The **length** of a string is the number of the terms in this string. The **empty string**, denoted by λ , is the string that has no terms.

Recurrence Relations

A **recurrence relation** for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer. A sequence is called a **solution** of a recurrence relation if its terms satisfy the recurrence relation.

The **initial conditions** for a recursively defined sequence specify the terms that precede the first term where the recurrence relation begins.

It can be shown that a recurrence relation together with its initial conditions determines a **unique solution**.

Fibonacci Sequence

The **Fibonacci Sequence**, $f_1, f_2, \dots, f_n, \dots$ is defined by the initial conditions $f_0 = 0, f_1 = 1$ and the recurrence relation

$$f_n = f_{n-1} + f_{n-2}$$

for $n = 2, 3, 4, \dots$

Find the Closed Formula

The explicit formula of the recurrence relation is called a **closed formula**. And we can use **iteration** to repeatedly use the recurrence relation. It can be done by forward substitution or backward substitution.

Summations

Introduction to Summations

The sum of

$$a_m, a_{m+1}, \dots, a_n$$

we can use the notation

$$\sum_{j=m}^n a_j$$

here j is called the **index of summation**, and m is **lower limit** and n is **upper limit**.

Summation of the Geometric Progressions

If a and r are real numbers and $r \neq 0$, then

$$\sum_{j=0}^n ar^j = \begin{cases} \frac{ar^{n+1}-a}{r-1} = \frac{a(1-r^{n+1})}{1-r} & \text{if } r \neq 1. \\ (n+1)a & \text{if } r = 1. \end{cases}$$

and it is called **geometric series**.

Some Useful Summation Formulae

Sum	Closed Form
$\sum_{k=0}^n ar^k (r \neq 1)$	$\frac{ar^{(n+1)} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^n k^3$	$\frac{n^2(n+1)^2}{6}$
$\sum_{k=0}^{\infty} x^k, \ x\ < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, \ x\ < 1$	$\frac{1}{(1-x)^2}$